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Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
MARCH 2025**

Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2020 Admission onwards)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

Answer any number of questions.

Each question carries 2 marks.

Maximum 25 Marks.

1. Define an analytic function. Give an example.
2. Define L'Hôpital Rule and using it compute $\lim_{z \rightarrow i} \frac{z^7 + i}{z^{14} + 1}$.
3. Verify the Cauchy-Riemann equations for $f(z) = \operatorname{Re}(z)$.
4. Define harmonic function and show that $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic.
5. Prove that the two families of curves $u(x, y) = x^2 - y^2 = c_1$ and $v(x, y) = 2xy = c_2$ form an orthogonal system.
6. Find the derivative of $e^{z^2 - (1+i)z + 3}$.
7. Find the value of $\cos i$ and $\sin(2+i)$.
8. Evaluate $\oint_C x \, dx$ where C is the circle defined by
 $x = \sin t, y = \cos t, 0 \leq t \leq 2\pi$.

Turn over

9. Using Cauchy's integral formula evaluate $\oint_C \frac{z^2 - 4z + 4}{z + i}$ where C is the circle $|z| = 2$.
10. State Cauchy's integral formula for derivatives.
11. State Ratio test.
12. State Taylor's theorem and write the Maclaurin series expression for $\sin z$.
13. State Laurent's theorem.
14. Prove that $f(z) = \frac{\sin z}{z}$ has a removable singular point.
15. State Cauchy's residue theorem.

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 35 Marks.

16. Show that the function $f(z) = x + 4iy$ is not differentiable at z .
17. Prove that an analytic function $f(z) = u + iv$ is constant if its (a) real part is constant ; and (b) its modulus is constant.
18. Briefly explain the exponential mapping $w = e^z$.
19. Find the values of $(1 + i)^i$.
20. Evaluate $\int_C \bar{z} dz$ where C is $x = 3t, y = t^2, -1 \leq t \leq 4$.
21. State and prove the fundamental theorem for contour integrals.

22. Find the radius of convergence of :

(a) $\sum_0^{\infty} \frac{z^n}{n!}$; and

(b) $\sum_1^{\infty} \frac{n! z^n}{n^n}$.

23. Find the Maclaurin series expansion of $f(z) = \frac{1}{(1-z)^2}$.

Section C

*Answer any **two** questions.*

Each question carries 10 marks.

Maximum 20 Marks.

24. State and prove Cauchy Riemann equations.

25. Expand $f(z) = \frac{1}{(z-1)^2(z-3)}$ in a Laurent series valid for

(a) $0 < |z-1| < 2$; and

(b) $0 < |z-3| < 2$.

26. State and prove :

(a) Fundamental theorem of algebra ; and

(b) Morera's theorem.

27. Evaluate $\int_0^{2\pi} \frac{1}{(2 + \cos \theta)^2} d\theta$.